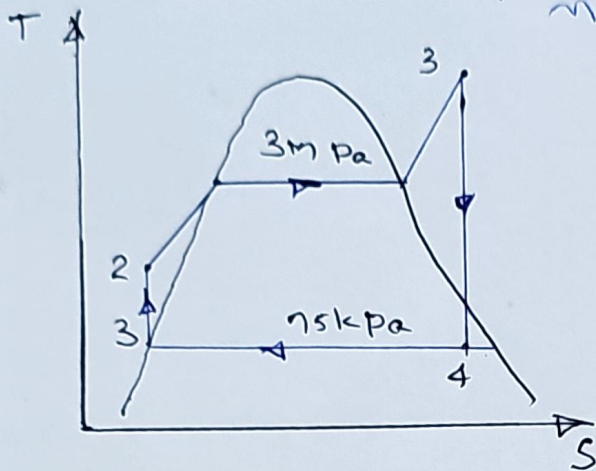


Example 1 The Simple Ideal Rankine cycle $\eta_{th} = ?$



Enthalpies at various point, using data from steam tables (Tables A-4, A-5, and A-6):

State 1: $P_1 = 75 \text{ kPa}$ } $h_1 = h_f @ 75 \text{ kPa} = 384.44 \text{ kJ/kg}$
 Sat. liquid } $v_1 = v_f @ 75 \text{ kPa} = 0.001037 \text{ m}^3/\text{kg}$

State 2: $P_2 = 3 \text{ MPa}$

$S_2 = S_1$

$w_{pump, in} = v_1 (P_2 - P_1) = (0.001037) (3000 - 75) \text{ kPa} \left(\frac{\text{kJ}}{\text{kPa} \cdot \text{m}^3} \right)$
 $= 3.03 \text{ kJ/kg}$

$h_2 = h_1 + w_{pump, in} = (384.44 + 3.03) \text{ kJ/kg} = 387.47 \text{ kJ/kg}$

State 3: $P_3 = 3 \text{ MPa}$ } $h_3 = 3116.1 \text{ kJ/kg}$
 $T_3 = 350^\circ\text{C}$ } $S_3 = 6.7450 \text{ kJ/kg} \cdot \text{K}$

State 4: $P_4 = 75 \text{ kPa}$ (sat. mixture)

$S_4 = S_3$

$x_4 = \frac{S_4 - S_f}{S_{fg}} = \frac{6.7450 - 1.2132}{6.2426} = 0.8861$

$h_4 = h_f + x_4 h_{fg} = 384.4 + 0.8861 (2278) = 2403 \text{ kJ/kg}$

Thus,

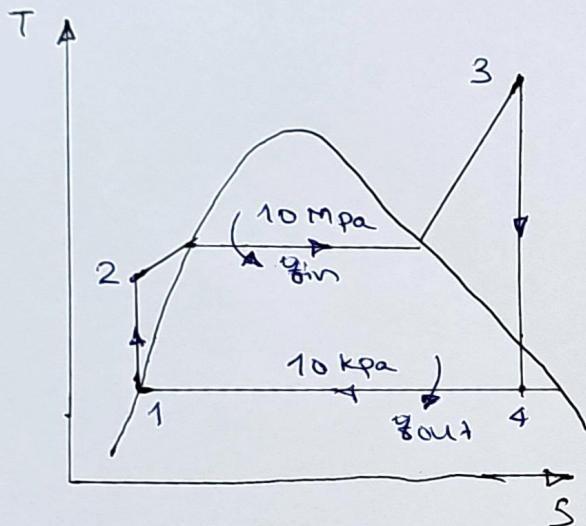
$q_{in} = h_3 - h_2 = (3116.1 - 387.47) \text{ kJ/kg} = 2728.6 \text{ kJ/kg}$

$q_{out} = h_4 - h_1 = (2403 - 384.44) \text{ kJ/kg} = 2018.6 \text{ kJ/kg}$

&

$\eta_{th} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{2018.6 \text{ kJ/kg}}{2728.6 \text{ kJ/kg}} = 0.26 \text{ or } 26\%$

Or
 $\eta_{th} = \frac{w_{net}}{q_{in}} = \frac{q_{in} - q_{out}}{q_{in}} = \frac{710 \text{ kJ/kg}}{2728.6} = 26\%$



$$\begin{aligned} X_4 &= ? \\ n+h &= ? \\ m &= ? \end{aligned}$$

Solution

- a) Use steam tables to find enthalpies at various points.
- State 1: $P_1 = 10 \text{ kPa}$ } $h_1 = h_f @ 10 \text{ kPa} = 191.81 \text{ kJ/kg}$
Sat. liquid } $v_1 = v_f @ 10 \text{ kPa} = 0.00101 \text{ m}^3/\text{kg}$

$$\begin{aligned} \text{State 2: } u_{1p, \text{in}} &= u_1(p_2 - v_1) \\ &= (0.00101 \text{ m}^3/\text{kg}) (10,000 - 10 \text{ kPa}) \left(\frac{1 \text{ kPa}}{1 \text{ kPa} \cdot \text{m}^3} \right) \\ &= 10.09 \text{ kJ/kg} \end{aligned}$$

$$h_2 = h_1 + w p_{\text{in}} = 191.81 + 10.09 = 201.90 \text{ kJ/kg}$$

$$\text{State 3: } \left. \begin{array}{l} P_3 = 10 \text{ MPa} \\ T_3 = 500^\circ\text{C} \end{array} \right\} \begin{array}{l} h_3 = 3375.1 \text{ kJ/kg} \\ s_3 = 6.5995 \text{ kJ/kg}\cdot\text{K} \end{array}$$

$$\left. \begin{array}{l} \text{state 4: } P_4 = 10 \text{ kPa} \\ S_4 = S_3 \end{array} \right\} x_4 = \frac{S_4 - S_f}{S_{fg}} = \frac{6.5995 - 0.6492}{7.4996} = 0.7934$$

$$h_4 = h_f + x_4 h_{fg}$$

$$= 191.81 + (0.7934)(2392.1) = 2089.7$$

$$b) \quad q_{in} = h_3 - h_2 = 3375.1 - 201.9 = 3173.2 \text{ kJ/kg}$$

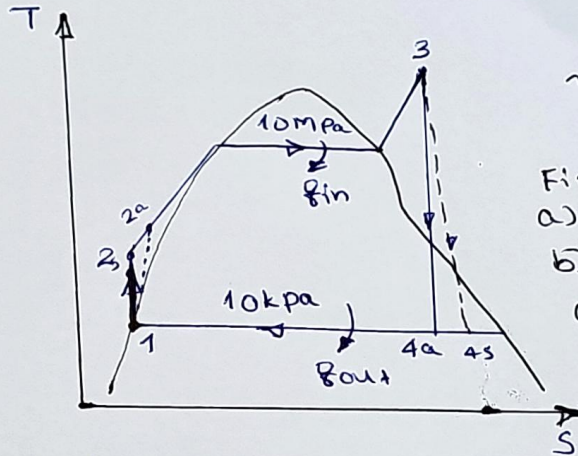
$$q_{out} = h_4 - h_1 = 2089.7 - 191.81 = 1897.9 \text{ kJ/kg}$$

$$W_{\text{net}} = \dot{Q}_{\text{in}} - \dot{Q}_{\text{out}} = 3173.2 - 1897.9 = 1275.4 \text{ kJ/kg}$$

$$\eta_{th} = \frac{W_{net}}{Q_{out}} = \frac{1275.4 \text{ kJ/kg}}{3173.2 \text{ kJ/kg}} = 40\%$$

$$c) \dot{m} = \frac{\dot{W}_{net}}{W_{net}} = \frac{210,000 \text{ kJ/s}}{1275.4 \text{ kJ/kg}} = 164.7 \text{ kg/s}$$

Example 3



$$\eta_T = 85\%$$

$$\eta_P = 85\%$$

Find
a) Quality

b) η_{th}

c) $\dot{m}$

Solution

a) From steam tables (Tables A-4, A-5 & A-6)

$$\text{State 1: } P_1 = 10 \text{ kPa} \left\{ \begin{array}{l} h_1 = h_f @ 10 \text{ kPa} = 191.81 \text{ kJ/kg} \\ \text{Sat. liquid} \end{array} \right. \left\{ \begin{array}{l} v_1 = v_f @ 10 \text{ kPa} = 0.0010 \text{ m}^3/\text{kg} \end{array} \right.$$

$$\text{State 2: } w_{P,in} = v_1 (P_2 - P_1) / \eta_P$$

$$= (0.00101 \text{ m}^3/\text{kg}) (10,000 - 10 \text{ kPa}) \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) / 0.85$$

$$= 11.87 \text{ kJ/kg}$$

$$h_2 = h_1 + w_{P,in} = 191.81 + 11.87 = 203.68 \text{ kJ/kg}$$

$$\text{State 3: } P_3 = 10 \text{ MPa} \left\{ \begin{array}{l} h_3 = 3375.1 \text{ kJ/kg} \\ T_3 = 500^\circ\text{C} \end{array} \right. \left\{ \begin{array}{l} s_3 = 6.5995 \text{ kJ/kg} \cdot \text{K} \end{array} \right.$$

$$\text{State 4: } P_4 = 10 \text{ kPa} \left\{ \begin{array}{l} x_{4s} = \frac{s_{4s} - s_f}{s_{fg}} = \frac{6.5995 - 0.6492}{7.4996} \\ s_{4s} = s_3 \end{array} \right.$$

$$= 0.7934$$

$$h_{4s} = h_f + x_{4s} h_{fg} = 191.81 + (0.7934)(2392.1) = 2089.7$$

$$\eta_T = \frac{h_3 - h_{4a}}{h_3 - h_{4s}} \rightarrow h_{4a} = h_3 - \eta_T (h_3 - h_{4s})$$

$$h_{4a} = 3375.1 - (0.85)(3375.1 - 2089.7)$$

$$= 2282.5 \text{ kJ/kg}$$

$$P_4 = 10 \text{ kPa} \left\{ \begin{array}{l} x_{4a} = \frac{h_{4a} - h_f}{h_{fg}} = \frac{2282.5 - 191.81}{2392.1} = 0.8 \end{array} \right.$$

$$h_{4a} = 2282.5$$

$$b) \dot{q}_{in} = h_3 - h_2 = 3375.1 - 203.68 = 3171.4 \text{ kJ/kg}$$

$$\dot{q}_{out} = h_4 - h_1 = 2282.5 - 191.81 = 2090.7 \text{ kJ/kg}$$

$$W_{net} = \dot{q}_{in} - \dot{q}_{out} = 3171.4 - 2090.7 = 1080.7 \text{ kJ/kg}$$

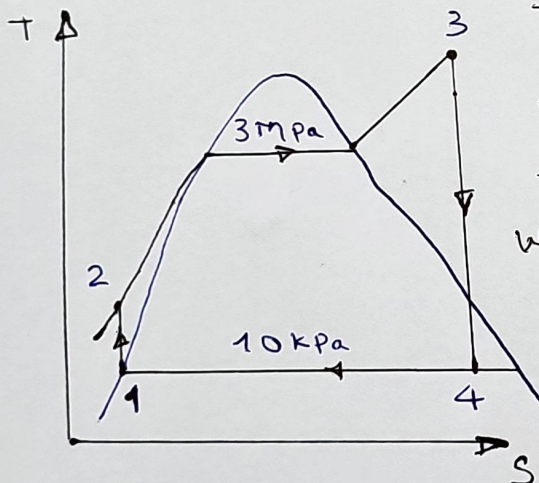
$$\eta_{th} = \frac{W_{net}}{\dot{q}_{in}} = \frac{1080.7 \text{ kJ/kg}}{3171.5 \text{ kJ/kg}} = 34.1\%$$

$$c) \dot{m} = \frac{\dot{W}_{net}}{W_{net}} = \frac{210,000 \text{ kJ/s}}{1080.7 \text{ kJ/kg}} = 194.3 \text{ kg/s}$$

EXAMPLE 4a) This is the steam power plant discussed in example 1.

$$P_1 = 10 \text{ kPa}, P_2 = P_3 = 3 \text{ MPa}, T_3 = 350^\circ\text{C} : \eta_{th} = ?$$

Solution



$$\text{State 1: } P_1 = 10 \text{ kPa} \left\{ \begin{array}{l} h_1 = h_f @ 10 \text{ kPa} \\ \text{Sat. liquid} \end{array} \right. \left\{ \begin{array}{l} = 191.8 \text{ kJ/kg} \\ v_1 = v_f @ 10 \text{ kPa} \end{array} \right.$$

$$v_1 = 0.00101 \text{ m}^3/\text{kg}$$

$$\text{State 2: } P_2 = 3 \text{ MPa} \\ s_2 = s_1$$

$$W_{p,in} = v_1 (P_2 - P_1) \\ = (0.00101 \text{ m}^3/\text{kg}) [(3000 - 10) \text{ kPa}] \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) \\ = 3.02 \text{ kJ/kg}$$

$$h_2 = h_1 + W_{p,in} = (191.81 + 3.02) \text{ kJ/kg} = 194.83 \text{ kJ/kg}$$

$$\text{State 3: } P_3 = 3 \text{ MPa} \left\{ \begin{array}{l} h_3 = 3116.1 \text{ kJ/kg} \\ T_3 = 350^\circ\text{C} \end{array} \right. \left\{ \begin{array}{l} s_3 = 6.7450 \text{ kJ/kg} \cdot \text{K} \end{array} \right.$$

$$\text{State 4: } P_4 = 10 \text{ kPa (sat. mixture)}$$

$$s_4 = s_3$$

$$x_4 = \frac{s_4 - s_f}{s_{fg}} = \frac{6.7450 - 0.6492}{7.4996} = 0.8128$$

$$h_4 = h_f + x_4 h_{fg} = 191.81 + 0.8128 (2392.1) = 2136.1 \text{ kJ/kg}$$

$$q_{in} = h_3 - h_2 = (3116.1 - 194.83) \text{ kJ/kg} = 2921.3 \text{ kJ/kg}$$

$$q_{out} = h_4 - h_1 = (2136.1 - 191.81) \text{ kJ/kg} = 1944.3 \text{ kJ/kg}$$

&

$$\eta_{th} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{1944.3 \text{ kJ/kg}}{2921.3 \text{ kJ/kg}} = 33.4\%$$

As a result of lowering the condenser pressure η_{th} increases from 26% to 33.4%. P from 75 kPa to 10 kPa.

4b) $P_3 = P_2 = 3 \text{ MPa}$, $T_3 = 600^\circ\text{C}$, State 4 (10 kPa & $s_4 = s_3$) are determined.

States 1 & 2 remain the same as 4a in this case.

$$h_3 = 3682.8 \text{ kJ/kg}$$

$$h_4 = 2380.3 \text{ kJ/kg} \quad (x_4 = 0.915)$$

Thus,

$$q_{in} = h_3 - h_2 = 3682.8 - 194.83 = 3488 \text{ kJ/kg}$$

$$q_{out} = h_4 - h_1 = 2380.3 - 191.81 = 2188.5 \text{ kJ/kg}$$

&

$$\eta_{th} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{2188.5 \text{ kJ/kg}}{3488 \text{ kJ/kg}} = 0.373 \text{ or } 37.3\%$$

Thermal efficiency increases from 33.4 to 37.3 percent as a result of superheating the steam from 350 to 600. At the same time, the quality of the steam increases from 81.3 to 91.5 percent. moisture content decreases from 18.7 to 8.5 percent.

c) State 1 remains the same in this case.

$$h_2 = 206.95 \text{ kJ/kg}$$

$$h_3 = 3583.1 \text{ kJ/kg}$$

$$h_4 = 2115.3 \text{ kJ/kg} \quad (x_4 = 0.804)$$

Thus,

$$q_{in} = h_3 - h_2 = 3583.1 - 206.95 = 3376.2 \text{ kJ/kg}$$

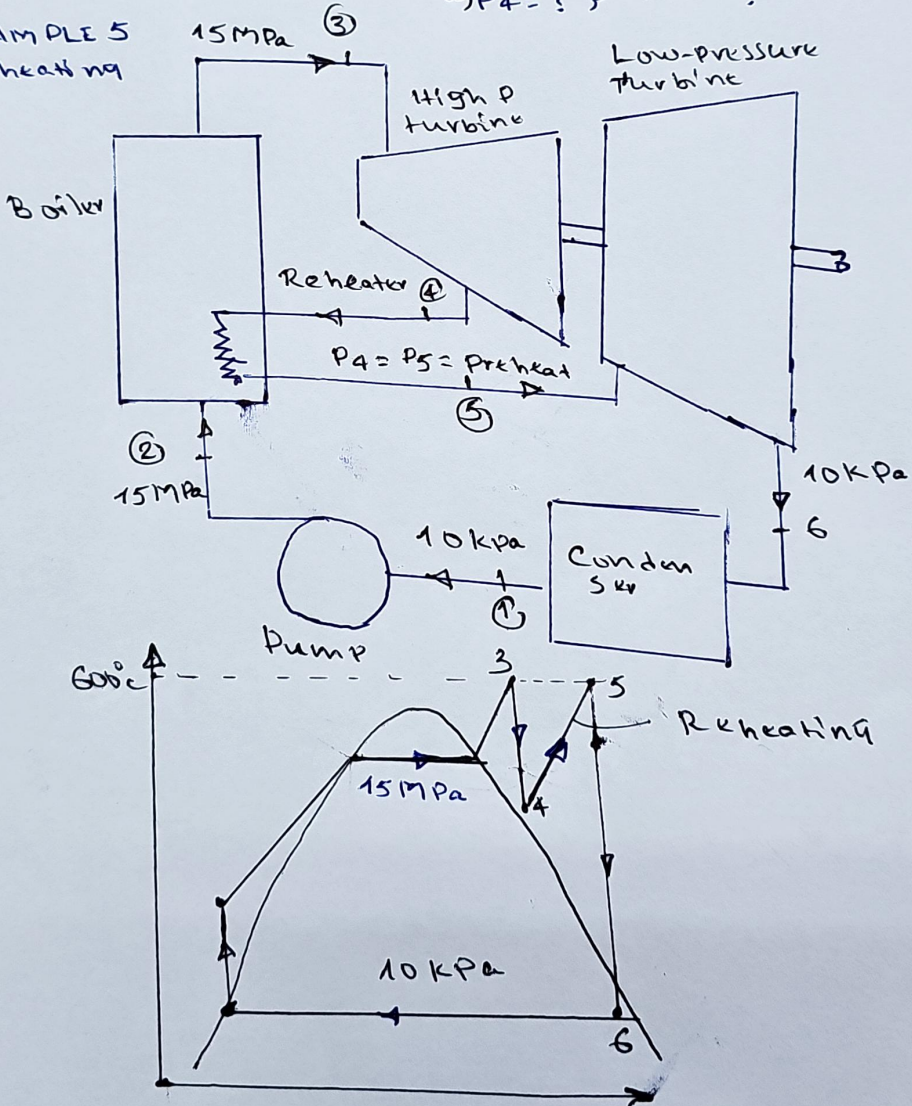
$$q_{out} = h_4 - h_1 = 2115.3 - 191.81 = 1923.5 \text{ kJ/kg}$$

$$\eta_{th} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{1923.5 \text{ kJ/kg}}{3376.2 \text{ kJ/kg}} = 0.43 \text{ or } 43\%$$

The thermal efficiency increases from 37.3 to 43 Percent as a result of raising the boiler pressure from 3 to 15 MPa while maintaining the turbine inlet temperature at 600°C. At the same time, however the quality of the steam decreases from 91.5 to 80.4 Percent (or the moisture content increases from 8.5 to 19.6 Percent)

a) $P_4 = ?$ b) $\eta_{th} = ?$

EXAMPLE 5 Reheating



a) The reheat pressure is determined from the condition that the entropies at states 4 & 6 be the same as the entropy at state 6:

$$P_6 = 10 \text{ kPa}$$

$$x_6 = 0.896 \text{ (Sat. mixture)}$$

$$s_6 = s_f + x_6 s_{fg} = 0.6492 + (0.896)(1.7262) = 2.161 \text{ kJ/kg} \cdot \text{K}$$

$$h_6 = h_f + x h_{fg} = 191.81 + 0.896(2392.1) = 2335.1 \text{ kJ/kg}$$

$$\left. \begin{array}{l} \text{State 5: } T_5 = 600^\circ\text{C} \\ S_5 = S_6 \end{array} \right\} \begin{array}{l} P_5 = 4 \text{ MPa} \\ h_5 = 3674.9 \text{ kJ/kg} \end{array}$$

Therefore, Steam should be reheated at a pressure of 4 MPa or lower to prevent a moisture content above 10.4%.

b) To determine the thermal efficiency, we need to know the enthalpies at all states:

$$\left. \begin{array}{l} \text{State 1: } P_1 = 10 \text{ kPa} \\ \text{Sat. liquid} \end{array} \right\} \begin{array}{l} h_1 = h_f @ 10 \text{ kPa} = 191.81 \text{ kJ/kg} \\ v_1 = v_f @ 10 \text{ kPa} = 0.00101 \text{ m}^3/\text{kg} \end{array}$$

$$\text{State 2: } P_2 = 15 \text{ MPa}$$

$$S_2 = S_1$$

$$w_{p,in} = v_1 (P_2 - P_1) = (0.00101 \text{ m}^3/\text{kg}) ([15,000 - 10] \text{ kPa}) \left[\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right]$$

$$= 15.14 \text{ kJ/kg}$$

$$h_2 = h_1 + w_{p,in} = (191.81 + 15.14) \text{ kJ/kg} = 206.95 \text{ kJ/kg}$$

$$\left. \begin{array}{l} \text{State 3: } P_3 = 15 \text{ MPa} \\ T_3 = 600^\circ\text{C} \end{array} \right\} \begin{array}{l} h_3 = 3583.1 \text{ kJ/kg} \\ S_3 = 6.6796 \text{ kJ/kg} \end{array}$$

$$\left. \begin{array}{l} \text{State 4: } P_4 = 4 \text{ MPa} \\ S_4 = S_3 \end{array} \right\} \begin{array}{l} h_4 = 3155 \text{ kJ/kg} \\ (T_4 = 375.5^\circ\text{C}) \end{array}$$

• Thus,

$$q_{in} = (h_3 - h_2) + (h_5 - h_4)$$

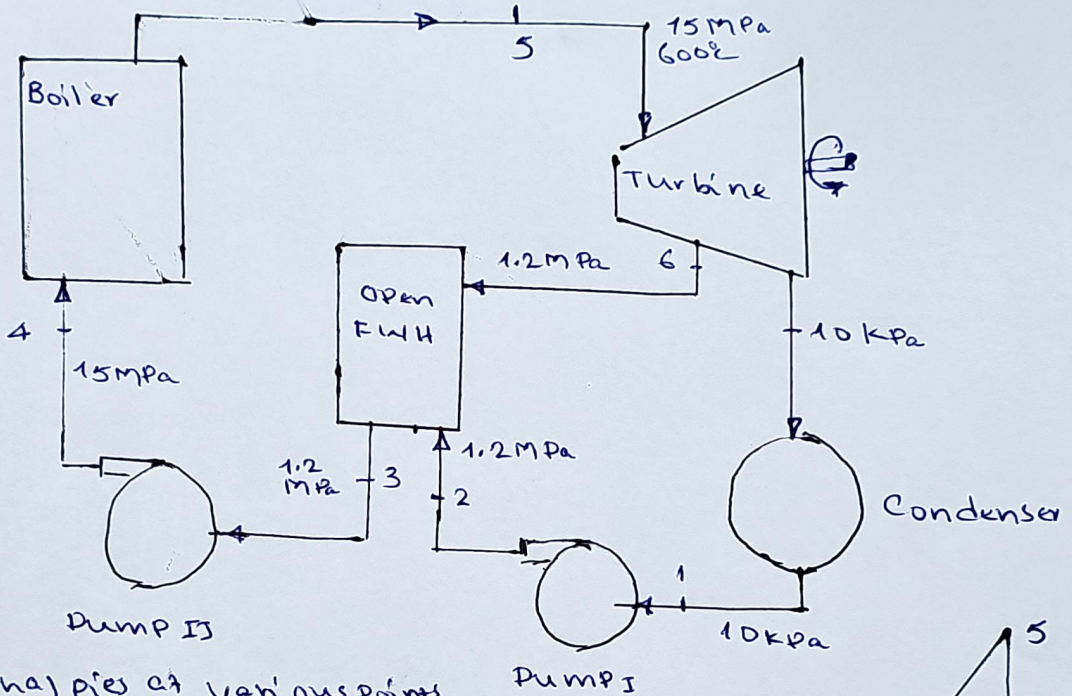
$$= (3583.1 - 206.95) \text{ kJ/kg} + (3674.9 - 3155) \text{ kJ/kg}$$

$$= 3896.1 \text{ kJ/kg}$$

$$q_{out} = h_6 - h_1 = (2335.1 - 191.81) \text{ kJ/kg} = 2143.3 \text{ kJ/kg}$$

$$\eta_{th} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{2143.3 \text{ kJ/kg}}{3896.1 \text{ kJ/kg}} = 0.45 \text{ or } 45\%$$

EXAMPLE 6 (Ideal Regenerative Rankine Cycle)



Enthalpies at various points

$$\text{State 1: } P_1 = 10 \text{ kPa} \left\{ \begin{array}{l} h_1 = h_{f@10 \text{ kPa}} \\ = 191.81 \\ v_1 = v_f@10 \text{ kPa} \\ = 0.00101 \text{ m}^3/\text{kg} \end{array} \right.$$

$$\text{State 2: } P_2 = 1.2 \text{ MPa} \left\{ \begin{array}{l} s_2 = s_1 \end{array} \right.$$

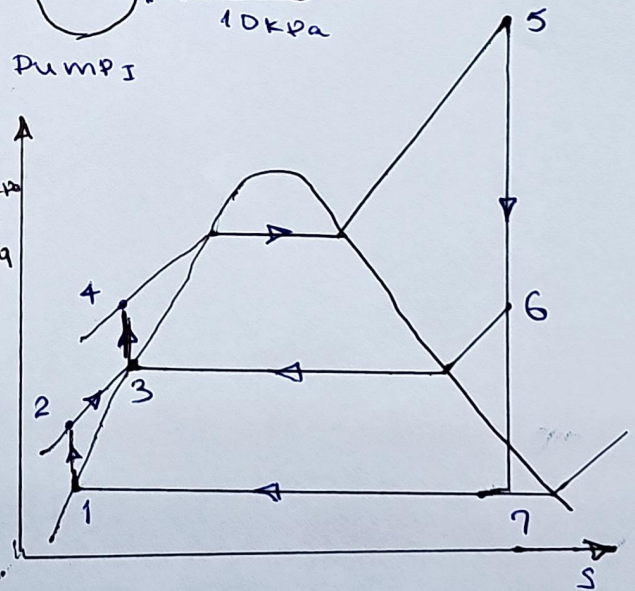
$$\begin{aligned} w_{P,I,\text{in}} &= v_1 (P_2 - P_1) \\ &= 0.00101 (1200 - 10) \text{ kJ/kg} \\ &= 1.2 \text{ kJ/kg} \end{aligned}$$

$$\begin{aligned} h_2 &= h_1 + w_{P,I,\text{in}} \\ &= (191.81 + 1.21) \text{ kJ/kg} = 193.01 \text{ kJ/kg} \end{aligned}$$

$$\text{State 3: } P_3 = 1.2 \text{ MPa} \left\{ \begin{array}{l} v_3 = v_f@1.2 \text{ MPa} = 0.001138 \text{ m}^3/\text{kg} \\ h_3 = h_f@1.2 \text{ MPa} = 798.33 \text{ kJ/kg} \end{array} \right.$$

$$\text{State 4: } P_4 = 15 \text{ MPa} \\ s_4 = s_3$$

$$\begin{aligned} w_{P,II,\text{in}} &= v_3 (P_4 - P_3) \\ &= (0.001138) (15,000 - 1200) \text{ kPa} \\ &= 15.7 \text{ kJ/kg} \end{aligned}$$



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$$h_4 = h_3 + W_{PI,in} = (798.33 + 15.70) \text{ kJ/kg} = 814.03 \text{ kJ/kg}$$

$$\text{State 5: } \left. \begin{array}{l} P_5 = 15 \text{ MPa} \\ T_5 = 600^\circ\text{C} \end{array} \right\} \begin{array}{l} h_5 = 3583.1 \text{ kJ/kg} \\ s_5 = 6.6796 \text{ kJ/kg}\cdot\text{K} \end{array}$$

$$\text{State 6: } \left. \begin{array}{l} P_6 = 1.2 \text{ MPa} \\ s_6 = s_5 \end{array} \right\} \begin{array}{l} h_6 = 2860.2 \text{ kJ/kg} \\ (T_6 = 218^\circ\text{C}) \end{array}$$

$$\text{State 7: } P_7 = 10 \text{ kPa}$$

$$x_7 = \frac{s_7 - s_f}{s_{fg}} = \frac{6.6796 - 0.6492}{7.4996} = 0.8041$$

$$h_7 = h_f + x_7 h_{fg} = 191.81 + 0.8041(2392.1) = 2115.3 \text{ kJ/kg}$$

Energy analysis of the mixing members

$$\dot{E}_{in} = \dot{E}_{out} \rightarrow \sum \dot{m} h_{in} = \sum \dot{m} h_{out}$$

or

$$y h_6 = (1-y) h_7 = 1(h_3)$$

where y is the fraction of steam extracted from the turbine ($= \dot{m}/\dot{m}_5$). Solving for y & substituting the enthalpy values, we find

$$y = \frac{h_3 - h_2}{h_6 - h_2} = \frac{798.33 - 193.01}{2860.2 - 193.01} = 0.2270$$

Thus,

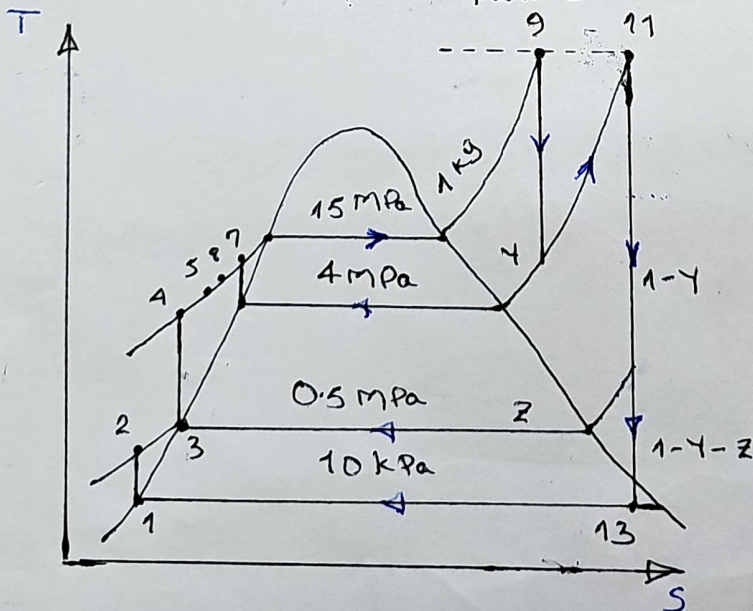
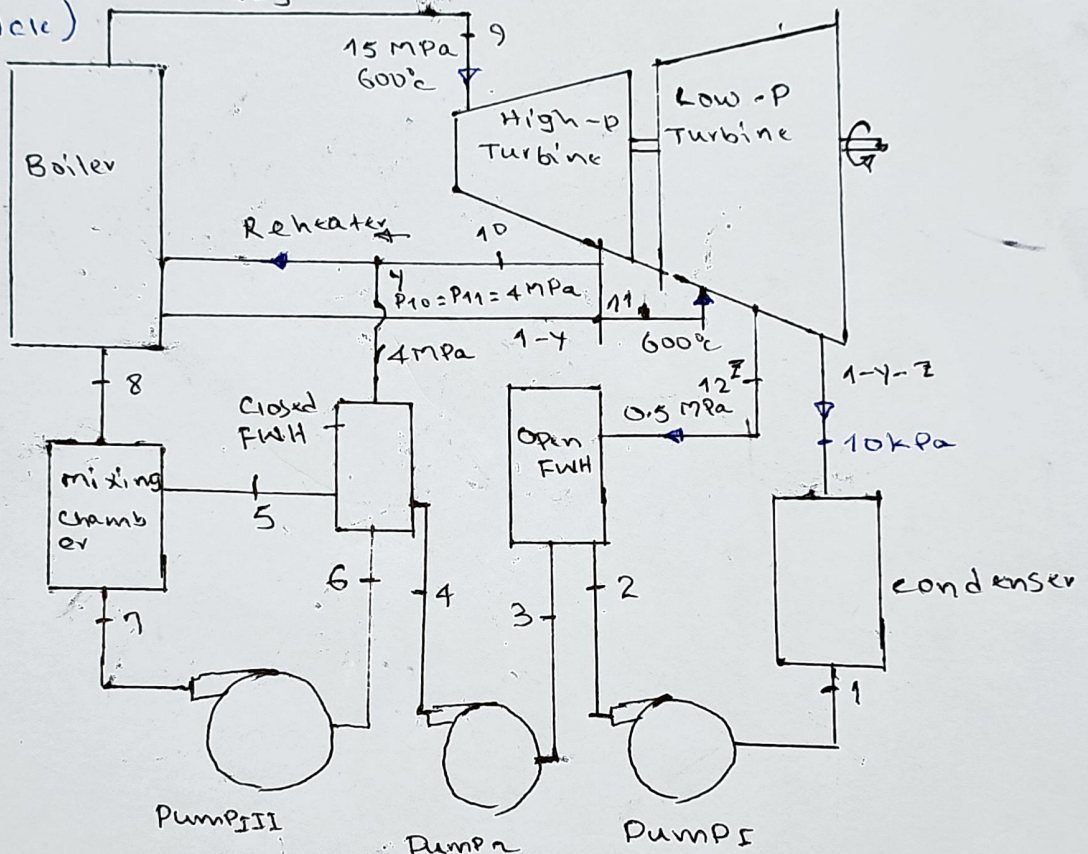
$$\dot{q}_{in} = h_3 - h_4 = (3583.1 - 814.03) \text{ kJ/kg} = 2769.1 \text{ kJ/kg}$$

$$\begin{aligned} \dot{q}_{out} &= (1-y)(h_7 - h_1) = (1 - 0.2270)(2115.3 - 191.81) \text{ kJ/kg} \\ &= 1486.9 \text{ kJ/kg} \end{aligned}$$

&

$$\eta_{th} = 1 - \frac{\dot{q}_{out}}{\dot{q}_{in}} = 1 - \frac{1486.9 \text{ kJ/kg}}{2769.1 \text{ kJ/kg}} = 0.43 \text{ or } 43\%$$

EXAMPLE 7 (The Ideal Reheat-Regenerative Rankine Cycle)



The enthalpies at the various state are:

$$h_1 = 191.81 \text{ kJ/kg}$$

$$h_2 = 192.30 \text{ kJ/kg}$$

$$h_3 = 640.09 \text{ kJ/kg}$$

$$h_4 = 643.92 \text{ kJ/kg}$$

$$h_5 = 1087.4 \text{ kJ/kg}$$

$$h_6 = 1087.4 \text{ kJ/kg}$$

$$h_7 = 1101.2 \text{ kJ/kg}$$

$$h_8 = 1089.8 \text{ kJ/kg}$$

$$h_9 = 3583.1 \text{ kJ/kg}$$

$$h_{10} = 3155.0 \text{ kJ/kg}$$

$$h_{11} = 3674.9 \text{ kJ/kg}$$

$$h_{12} = 3014.8 \text{ kJ/kg}$$

$$h_{13} = 2335.7 \text{ kJ/kg}$$

$$w_{PI,in} = 0.49 \text{ kJ/kg}$$

$$w_{PII,in} = 3.83 \text{ kJ/kg}$$

$$w_{PIII,in} = 13.77 \text{ kJ/kg}$$

The fractions of steam extracted are determined from the mass & energy balances of the feedwater heaters. Closed feedwater heater:

$$\dot{E}_{in} = \dot{E}_{out}$$

$$y h_{10} + (1-y) h_4 = (1-y) h_5 + y h_6$$

$$y = \frac{h_5 - h_6}{(h_{10} - h_6) + (h_5 - h_4)} = \frac{1087.4 - 643.92}{(3155 - 1087.4) + (1087.4 - 643.92)}$$

$$= 0.1766$$

Open feed water heater:

$$\dot{E}_{in} = \dot{E}_{out}$$

$$z h_{12} + (1-y-z) h_2 = (1-y) h_3$$

$$z = \frac{(1-y)(h_3 - h_2)}{h_{12} - h_2} = \frac{(1-0.1766)(640.09 - 192.30)}{3014.8 - 192.30} = 0.1306$$

The enthalpy at 8 is determined from/by applying the mass & energy equations to the mixing chamber, which is assumed to be insulated:

$$\dot{E}_{in} = \dot{E}_{out}$$

$$(1) h_8 = (1-y) h_5 + y h_7$$

$$h_8 = (1-0.1766)(1087.4) \text{ kJ/kg} + 0.1766(1101.2) \text{ kJ/kg}$$

$$= 1089.8 \text{ kJ/kg}$$

Thus,

$$\begin{aligned}
 \dot{q}_{in} &= (h_9 - h_8) + (1-y)(h_{11} - h_{10}) \\
 &= (3583.1 - 1089.8) \text{ kJ/kg} + (1 - 0.1766)(3674.9 - 3155) \text{ kJ/kg} \\
 &= 2921.4 \text{ kJ/kg}
 \end{aligned}$$

$$\begin{aligned}
 \dot{q}_{out} &= (1-y-z)(h_{13} - h_1) \\
 &= (1 - 0.1766 - 0.1306)(2335.7 - 191.81) \text{ kJ/kg} \\
 &= 1485.3 \text{ kJ/kg}
 \end{aligned}$$

Q

$$\eta_{th} = 1 - \frac{\dot{q}_{out}}{\dot{q}_{in}} = 1 - \frac{1485.3 \text{ kJ/kg}}{2921.4 \text{ kJ/kg}} = \underline{0.4920 \text{ or } 49.2\%}$$